Pixel-Wise Semisupervised Fabric Defect Detection Method Combined With Multitask Mean Teacher

Linhao Shao, Erhu Zhang, Qiurui Ma, and Mei Li

***Abstract*— Fabric defect detection is a crucial problem in the production of textile industry. Nowadays, a large number of fabric defect detection algorithms based on deep learning have been proved to have excellent detection results, but the performance of detection often depends on large-scale labeled datasets. However, it is very laborious and costly to label a large number of pixel-level defect samples in industrial production. Although semisupervised detection method can reduce the labeling workload, fabric defect detection is still a challenging task due to the diversity of defects and the low contrast between defects and background. To address these issues, this article proposes a pixel-wise semisupervised fabric defect detection method combined with multitask mean teacher (MT). We focus on establishing a multitask detection model for defect area detection, defect contour detection, and defect distance map detection by leveraging their auxiliary information in better defect segmentation and assign this model as a student network and teacher network. Finally, the multitask supervised loss for labeled images and the multitask consistency loss for unlabeled images to train the network. Experiments on three public fabric defect detection datasets show that our method outperforms the current mainstream pixel-wise segmentation methods with MT, which verifies that the proposed network can effectively detect fabric defects by a small number of labeled samples.**

***Index Terms*— Fabric defect detection, mean teacher (MT) network, multitask learning, pixel-wise segmentation, semisupervised learning.**

I. Introduction

FABRIC defect detection plays an important role in textile industry. Traditional fabric defect detection methods mainly rely on human eyes to distinguish defective products, which is time-consuming and has a high false detection rate. Therefore, manual detection cannot meet the actual production needs. There is an urgent need to develop automatic fabric defect detection methods to deal with the textile industry production.

The traditional fabric defect detection methods can be divided into four categories: statistical-based methods [1], frequency domain analysis methods [2], model-based methods [3], and low-rank decomposition methods [4]. These traditional fabric defect detection methods have made a lot of contributions to the development of this field. Among them, reference [1] is the basic work of structural defect identification, which helps us understand the essence of structural defects. Reference [2] is the basic work of analyzing images with wavelet transform in frequency domain, which provides a reference for fabric defect detection using frequency domain analysis in the later stage. Zachevsky and Zeevi [3] propose a model based on the combination of direction and consistency features and color natural stochastic textures (NSTs), which is the basic work of model-based method. Mo *et al.* [4] propose an effective method based on low-rank decomposition to improve the accuracy of defect detection and transform the problem of fabric defect detection into the optimization problem between fabric image and defect prediction. As some basic work, the above methods provide a lot of ideas for the development of fabric defect detection. However, these traditional methods mainly use manual feature extraction, which cannot adaptively select features to detect defects in complex scenes. Recently, deep convolutional neural network (DCNN) has achieved excellent performance in the fields of image recognition [5], image segmentation [6], object detection [7], and so on. Therefore, in recent years, many researchers have adopted CNN to fabric defect detection and achieved satisfying results. Experiments [8], [9] show that the algorithm outperforms other traditional algorithms under the training of large industrial datasets. Compared with the traditional fabric defect detection methods, deep learning can extract richer image features and has robust adaptability and detection accuracy.

The fabric defect detection based on deep learning has achieved good detection results, but these methods still have the following two problems.

1) These methods rely on large-scale annotated defect image datasets. In the actual textile industry production, it is very laborious to obtain large-scale labeled training set. Especially due to the diversity of fabrics, even in an industrial production line, more than 200 types of fabric defects may be produced, so the annotation of these defect images is a great difficulty. Therefore, it also limits the full application of deep learning in textile manufacturing industry.
2) These network models ignore the value of prior information of the defect image itself. At present, most pixel-wise segmentation networks (SegNets) are based on classical encoder–decoder network structures, such as early network structures fully convolutional network (FCN) [6] and U-shaped network(UNet) [10], as well as networks bilateral segmentation network (BiSeNet) [11], UNet++

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[12], and short-term dense concatenate network (STDC) [13] in recent years. The design of these network structures is constantly improving the performance of semantic segmentation. However, these encoder–decoder networks suffer from the loss of structural information due to the downsampling operations via the max-pooling layer [14], which is not conducive to fabric defect detection at the pixel-wise. How to use the prior information of the defect image itself to make up for the lost structural information is another difficulty. These difficulties motivate us to explore a better deep network model to solve the problem of fabric defect detection under a small number of annotated samples.

For the first problem, we propose a semisupervised model to solve the problem of fabric defect detection under a small number of labeled samples. In the actual textile industry production, it is easier for us to obtain a large number of unlabeled actual defect image and use a part of labeled samples and unlabeled samples as the training set, which can save a lot of annotation cost. In recent years, mean teacher (MT) [15] structure is the mainstream semisupervised network model. Inspired by this model, we construct a pixel-wise semisupervised segmentation method based on MT network. For the second problem, it is found through experiments that under a small amount of labeled data, it is difficult to achieve an ideal defect detection effect only by using the existing MT network. This is because in semisupervised learning, there is only a small part of labeled data, and the network can only obtain limited defect features. Therefore, we pay more attention to the design of teacher network and student network. In order to make the network fully obtain defect features, the medical image segmentation of deep contour-aware network (DCAN) [16] and deep multi-task network (DMTN) [17] has aroused our interest. In the case of DCAN, the learning of segmentation mask is guided by contour information, and in DMTN, the learning of segmentation mask is guided by distance map (DM). Inspired by these two works, we rethink the problem of structural information loss of defect detection network. Fortunately, the contour information and DM are just the structural prior information of the defect area (as shown in Fig. 1). The defect contour and DM are used as structural information to compensate the loss of structural information due to the max-pooling layer. In addition, contour detection and DM detection can be used as auxiliary tasks to help the network segment the defect area better. In order to better coordinate several tasks, inspired from the multitask learning work [18], we design a semisupervised network combined with multitask learning to solve the problem of fabric defect detection.

Based on the above analysis, we propose a semisupervised multitask MT network to boost the performance of fabric defect detection. We first focus on designing a multitask student and teacher network, denoted as ST-CNN, which is used to learn three tasks: defect segmentation detection, defect contour detection, and defect DM detection. Then, according to MT network, we propose multitask supervised loss for labeled images (LIs) and multitask consistency loss to ensure the consistency of student network and teacher network for unlabeled images (UIs). Training the model through multitask supervised loss and multitask consistency loss, our network can accurately detect common fabric defects. This article constructs experiments on three public fabric defect detection datasets. The experimental results show that our network can achieve general detection results and achieve the state-of-the-art detection results compared with other mainstream segmentation methods with MT.

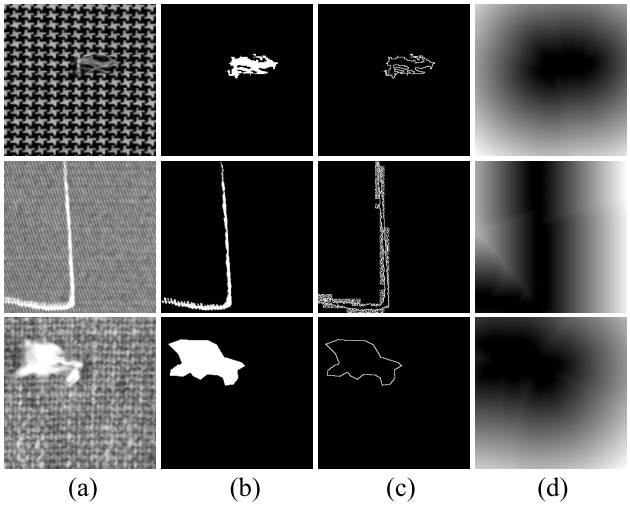


Fig. 1. Fabric defects and their structural prior information. (a) Defect image. (b) Mask. (c) Contour. (d) DM.

In summary, three major contributions are as follows.

1) A pixel-wise semisupervised fabric defect detection network is proposed, which has achieved the state-of-the-art performance results on three general real-world fabric defect detection datasets.
2) A multitask learning structure (ST-CNN) combined with MT network is proposed. More specifically, the defect contour information and DM are incorporated into the fabric defect detection network as structural prior information for the first time, which realizes more accurate fabric defect detection.
3) The multitask supervised loss function and multitask consistency loss function along with deep supervision are proposed to optimize the network and improve the detection accuracy.

II. Related Work

In the past decades, many researchers have proposed a large number of methods for fabric defect detection. These methods can be roughly divided into traditional methods and learning-based methods. In this section, we will review the existing fabric defect detection methods and their related technologies through two parts: 1) traditional fabric defect detection methods and 2) deep-learning-based fabric defect detection methods.

A. Traditional Fabric Defect Detection

The most common traditional fabric defect detection methods can be divided into four categories: statistical-based methods [1], frequency domain analysis methods [2], model-based methods [3], and low-rank decomposition methods [4]. The statistical-based method detects defects according to the spatial distribution difference of gray values between defect region

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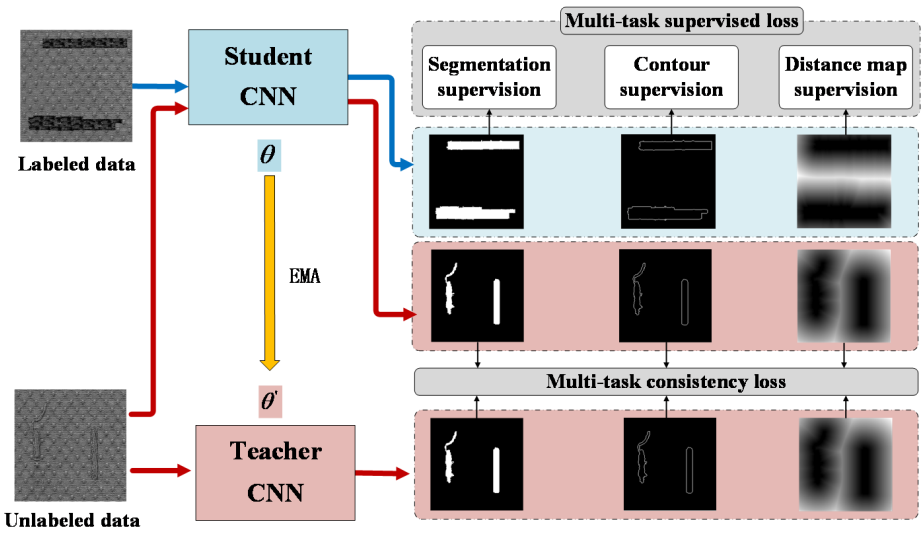


Fig. 2. Architecture of the proposed fabric defect detection network.

and nondefect region, which mainly include autocorrelation analysis, gray-level co-occurrence matrix (GLCM), morphological analysis, and fractal dimension features. The method based on frequency domain analysis is to convert the fabric image from spatial domain to frequency domain and then analyze and process the fabric image in frequency domain, which mainly includes Fourier transform [19], wavelet transform [20], Gabor transform [21], and so on. The model-based method is to construct the model and parameter estimation and extract the texture features of the fabric image. It is suitable for the defect image of fabric surface changes caused by yarn breakage, needle breakage, and other defects. The existing methods include the Gaussian mixture model (GMM) [22] and Gaussian Markov random field (GMRF) [23]. These methods achieve satisfactory detection results, but it has high computational complexity and insufficient detection of small defects. In addition, low rank decomposition methods rely on effective descriptors and also require huge amount of computations. The above traditional methods have made great contributions to fabric defect detection, but the traditional methods need to extract features manually, and they need to be further developed in detection accuracy and time efficiency.

B. Deep-Learning-Based Fabric Defect Detection

In the past few years, deep learning has achieved great success in the fields of image recognition, image segmentation, and object detection. Many researchers have applied deep learning technology to defect detection and achieved satisfactory results. Deep-learning-based fabric defect detection methods can be divided into three categories: 1) classification network-based methods; 2) detection network-based methods; and 3) SegNet-based methods. For classification network-based model, Wenkang *et al.* [24] proposed a CNN classification network for on-loom fabric defect detection. Zhao *et al.* [25] proposed a CNN classification model based on visual long short-term memory (VLSTM). However, the above methods can only classify the defect image at the image level and cannot locate the specific defect area. For detection network-based model, it is mainly divided into one-stage detection algorithm [9], [26] and two-stage detection algorithm [27]. These one-stage detection algorithms have fast detection speed, but the detection accuracy usually cannot meet the actual requirements. On the contrary, the two-stage methods have higher detection accuracy, but sometimes, the detection speed is difficult to meet the needs of the actual industrial scenes. The above detection-based method can roughly locate the defect area, and it cannot obtain accurate information of defects. For SegNet-based model, Huang *et al.* [28] designed a two-stage segmentation and decision network and achieved good fabric defect detection results. This method achieves pixel-level fabric defect detection and can obtain accurate defect information, but in the training of these networks, a large amount of data and paired datasets are needed for supervised learning. However, in the actual fabric defect detection environment, the defect sample is relatively few, and it is difficult to label defects. Based on the above problems, in this article, we propose a semisupervised defect detection model with few labeled defect samples, which can achieve better fabric defect detection results compared with other methods.

III. Methodology

A. System Overview

The design of the overall network architecture in Fig. 2 is based on semisupervised MT [15]. The core idea of this method is that the model acts as both a student and a teacher. As a teacher, it is used to generate students' learning goals. As a student, it uses the goals generated by the teacher model to learn. On the basis of this theory, we design a multitask learning MT network for fabric defect detection task. The core is to use the structural prior information of defect area (defect contour and defect DM) as auxiliary information to guide the more refined segmentation of the network. The overall structure includes two identical networks: student CNN and teacher CNN.

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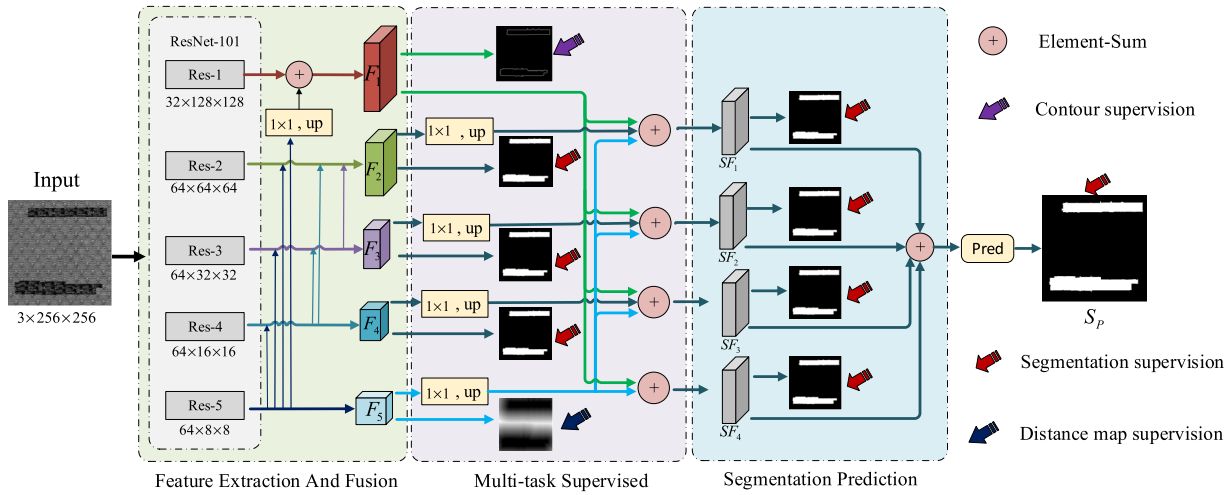


Fig. 3. Architecture of student CNN and teacher CNN (ST-CNN).

1) *Student Network:* The input of student network is labeled defect images and unlabeled defect images. When a labeled defect image is sent to the student network, its predicted contour map, DM, and segmentation map and their corresponding ground truth construct a multitask supervised loss function. When an unlabeled defect image is sent to the student network, the prediction results of student network and teacher network construct a multitask consistency loss function, which mainly ensures that the prediction results of the teacher network are as similar as the prediction labels of the student network as possible. The parameters of the whole student network are updated through two parts loss function.
2) *Teacher Network:* The parameters of teacher network are updated through the moving average of the network parameters of student model. After repeated iterative training and parameter updating, the student network for fabric defect detection can be obtained. In the test phase, we only use the student CNN to predict the input defect image.

B. Multitask Detection Network

1) Feature Extraction and Fusion: Fig. 3 shows our proposed multitask detection network ST-CNN, that is, the student CNN and teacher CNN in Fig. 2. First, given an input fabric defect image, we use the pretrained model ResNeXt-101 [29] to generate five-level feature maps, which are Res-1, Res-2, Res-3, Res-4, and Res-5, respectively (see the left of Fig. 3). Then, as Jarrett *et al.* [30] indicate, the different layers of CNN contain different feature information. In fabric defect detection, shallow features generally contain more contour information of defect area, and deep features generally contain more global semantic information of defect area. Therefore, in order to effectively segment the defect area, we divide the five levels of features into low-level features (Res-1), middle-level features (Res-2, Res-3, and Res-4), and high-level features (Res-5). The low-level features are used for defect contour detection, high-level features for defect DM detection, and middle-level features for defect segmentation. Meanwhile, in the classical FCN network [6], we can see the higher level features, which contain more global context information. Therefore, in the design of feature fusion, we integrate the high-level features into the middle-level and low-level features, which is more conducive to defect contour detection and segmentation detection. However, in fabric defect detection, the structure of the network cannot be too complex to better apply to the actual industrial scene. Based on this consideration, a lightweight feature fusion method is proposed, which gradually concatenates the high-level features to the low-level features, as shown in Fig. 3, and finally obtains the fused features $F_i (i = 1, \ldots, 5)$.

2) Defect Contour and DM Detection: In the current fabric defect segmentation methods, by observing the segmentation results, we notice that the boundaries of the defect region are difficult to distinguish from the surrounding nondefect region. This motivates us to think about using contour information to constrain the defect area and better enhance the detection performance. In our ST-CNN, we fuse shallow CNN feature (Res-1) and deep CNN feature (Res-5) to generate F_1 feature map, which is used to predict the contour of fabric defects. It can be expressed by the following formula:

$$F_1 = \text{Res-1} + \text{up}(\text{Conv}_{1\times1}(\text{Res-5})) \tag{1}$$

where $up()$ represents upsampling. In Section III-B1, we analyze that the shallow layer contains more detailed information, but the shallow layer also contains a lot of noise interference information of nondefect areas, so only the shallow features cannot fully obtain the contour features. Therefore, we use the deep feature (Res-5) to supplement the shallow feature, which has a larger receptive field to effectively bypass the nondefective pixels. Finally, the contour feature F_1 can be computed via an element-wise addition on Res-1 and Res-5.

In our ST-CNN, the DM represents the global priori structural information of the defect area, while in CNN, the features extracted from the high level are also global semantic features. In order to simplify the design of the network, Res-5 of the fifth layer is directly used to predict the DM information.

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3) Defect Segmentation Prediction: To effectively use the prior structure information of contour and DM for assisting defect segmentation, we integrate the shallow features supervised by contour information and deep features supervised by DM into the middle layer to obtain defect segmentation features. As shown in Fig. 3, $F_i (i = 2, 3, 4)$ represents the middle layer feature, and F_1 and F_5 represent the shallow layer feature and deep layer feature, respectively. The final segmentation features $\mathrm{SF}_j (j = 1, \ldots, 4)$ can be obtained by adding the shallow layer feature and deep layer feature to the middle layer feature, which are expressed as the following formula:

$$\mathrm{SF}_j = \mathrm{up}(\mathrm{Conv}_{1\times 1}(F_i)) + F_1 + F_5 \tag{2}$$

where up() represents upsampling. After obtaining the segmentation features SF_j, all features are element-wise added together to obtain the final prediction result S_P, which can be expressed as follows:

$$S_\mathrm{P} = \mathrm{pred}\left(\sum_{j=1}^{4} \mathrm{SF}_j\right) \tag{3}$$

where pred indicates the use of three 3×3 convolutional layers, a 1×1 convolutional layer, and a sigmoid activation layer on the features.

4) Deep Supervision: In this article, we design a deep supervision method to make full use of the priori structure information of contour and DM to assist image defect segmentation. As shown in Fig. 3, the shallow prediction result is supervised by the labeled defect contour image (CI), the deep prediction result is supervised by the labeled DM, and the defect segmentation prediction results are supervised by the labeled defect images by two-level supervision. This two-level deep supervision is more conducive to the network to learn more appropriate parameters.

C. Multitask Loss Function

1) Multitask Supervised Loss: For labeled fabric defect data, we have paired input defect images and corresponding annotated segmentation mask. Naturally, we take the labeled segmentation mask as the ground truth of defect segmentation prediction (G_s). Then, the edge detection algorithm is used to generate the defect contour on labeled segmentation mask as the ground truth of defect contour detection (G_c). The DM is generated on the labeled segmentation mask as the ground truth for defect DM detection (G_d).

After obtaining the ground truth of defect image, defect contour, and defect DM, as shown in Fig. 2, the multitask supervised loss (L_s) for a labeled fabric defect image (x) is calculated by adding the supervised losses of the defect segmentation detection (L_s^s), defect contour detection (L_s^c), and defect DM detection (L_s^d), i.e.,

$$L_\mathrm{s}(x) = L_\mathrm{s}^\mathrm{s} + \alpha L_\mathrm{s}^\mathrm{c} + \beta L_\mathrm{s}^\mathrm{d} \tag{4}$$

where

$$\begin{aligned} L_\mathrm{s}^\mathrm{s} &= \sum_{i=1}^{8} \Phi_\mathrm{BCE}(P_\mathrm{s}(i), G_\mathrm{s}) \\ L_\mathrm{s}^\mathrm{c} &= \Phi_\mathrm{BCE}(P_\mathrm{c}, G_\mathrm{c}) \\ L_\mathrm{s}^\mathrm{d} &= \Phi_\mathrm{BCE}(P_\mathrm{d}, G_\mathrm{d}) \end{aligned} \tag{5}$$

where $P_\mathrm{s}(i)$ represents the eight predicted defect segmentation image (the red arrow in Fig. 3), P_c represents the predicted contour (the purple arrow in Fig. 3), and P_d represents the predicted DM (the blue arrow in Fig. 3). $\Phi_\mathrm{BCE}(.)$ is the binary cross-entropy loss function. We set the weight coefficients $\alpha = 10$ and $\beta = 10$ in the training of the network.

2) Multitask Consistency Loss: For the unlabeled fabric defect data, we put the unlabeled data (y) into the ST-CNN to obtain the results of three tasks:

1) in the student network: defect segmentation map (S_s), defect contour map (S_c), and defect DM (S_d);
2) in the teacher network: defect segmentation map (T_s), defect contour map (T_c), and defect DM (T_d).

Then, we make the prediction results of the three tasks from teacher network and the student network to be consistent, resulting in a multitask consistency loss (L_c), which is expressed as follows:

$$L_\mathrm{c}(y) = L_\mathrm{c}^\mathrm{s} + L_\mathrm{c}^\mathrm{c} + L_\mathrm{c}^\mathrm{d} \tag{6}$$

where

$$\begin{aligned} L_\mathrm{c}^\mathrm{s} &= \sum_{j=1}^{8} \Phi_\mathrm{MSE}(S_\mathrm{s}(j), T_\mathrm{s}(j)) \\ L_\mathrm{c}^\mathrm{c} &= \Phi_\mathrm{MSE}(S_\mathrm{c}, T_\mathrm{c}) \\ L_\mathrm{c}^\mathrm{d} &= \Phi_\mathrm{MSE}(S_\mathrm{d}, T_\mathrm{d}) \end{aligned} \tag{7}$$

where L_c^s, L_c^c, and L_c^d refer to the consistency loss of the fabric defect segmentation, defect contour detection, and defect DM detection, respectively, and $\Phi_\mathrm{MSE}(.)$ is the mean square error (MSE) loss function.

3) Total Loss of Network: The total loss function of the network is defined as the weighted sum of the multitask supervised loss and the multitask consistency loss, as shown in the following formula:

$$L_\mathrm{total} = \sum_{i=1}^{M} L_\mathrm{s}(x_i) + \lambda \sum_{j=1}^{N} L_\mathrm{c}(y_j) \tag{8}$$

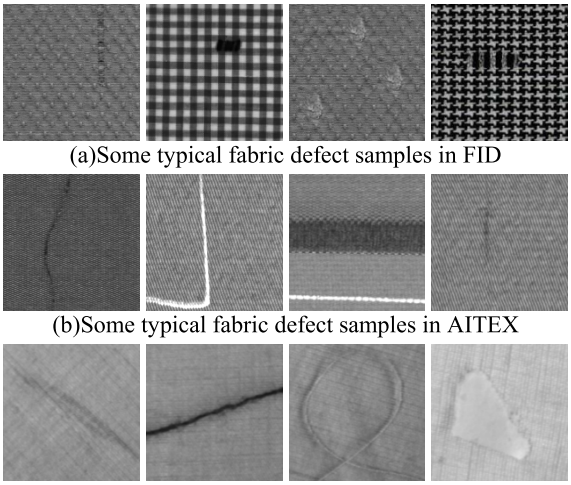
where M and N refer to the number of labeled fabric defect images and unlabeled fabric defect images in the training set, respectively, $L_\mathrm{s}(x_i)$ denotes the multitask supervised loss [see (4)] for the ith labeled defect image, and $L_\mathrm{c}(y_j)$ denotes the multitask consistency loss [see (6)] for the jth unlabeled defect image. The weight λ is used to balance the multitask supervised loss and multitask consistency loss. Following [31], λ can be updated using a time-dependent Gaussian warming up function: $\lambda(t) = \lambda_\mathrm{max} e^{(-5(1-t/t_\mathrm{max})^2)}$, where t refers to the current training iteration and t_max refers to the maximum training iteration. In our experiments, we initialize λ_max to 10. The training process of the whole network is the same as that of the classical MT network, refer to [15].

IV. Experiment and Discussion

A. Experimental Datasets

In our experiment, three public fabric defect datasets are selected to comprehensively evaluate our methods, which are fabric image dataset (FID) [32], AITEX [33], and TILDA [34].

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(a)Some typical fabric defect samples in FID

(b)Some typical fabric defect samples in AITEX

(c)Some typical fabric defect samples in TILDA

Fig. 4. Some typical fabric defect images in actual production. (a) Some typical fabric defect samples in FID. (b) Some typical fabric defect samples in AITEX. (c) Some typical fabric defect samples in TILDA.

TABLE I

NUMBER OF TRAINING SETS AND TEST SETS SELECTED IN THE EXPERIMENT. LIS, UIS, CIS, AND DM

Dataset	Train		Test
FID	LI:170	UI:170	44
	CI:170	DM:170	
AITEX	LI:515	UI:515	100
	CI:515	DM:515	
TILDA	LI:320	UI:320	160
	CI:320	DM:320	

1) FID Datasets: The FID dataset is provided by the University of Hong Kong, Hong Kong. It includes star fabric, box fabric, and dot fabric, and each fabric contains 30 pieces of 256×256 normal images [Fig. 4(a)]. In our experiment, we first expand the original dataset through general data augmentation methods, such as rotation, folding, and so on. Then, we divided the original dataset into training set and test set. The training set includes labeled defect images, unlabeled defect images, defect CIs, and defect DM (as shown in Table I).

2) AITEX Datasets: The AITEX dataset contains common different types of fabric defects [Fig. 4(b)] produced in the production of textile industry. There are 245 images of 4096×256 pixels and their corresponding segmentation mask. In our experiment, we first crop the defect image of the dataset to the size of 256×256 and then divide the dataset into training set and test set (as shown in Table I).

3) TILDA Datasets: The TILDA dataset mainly includes twill or plain fabric images with simple textures [Fig. 4(c)], such as multiple netting, holes, broken ends, thin bars, and thick bars. We select some typical defect images and crop them to a size of 256×256. The dataset is divided into training set and test set (as shown in Table I).

B. Experimental Settings

1) Training Phase: In the training stage, when designing ST-CNN, in order to fully extract the features of defect images, we select ResNeXt-101 [29] as the backbone network to extract features, which has been well pretrained in ImageNet. Therefore, the initial parameters of the network are the pretraining model parameters of ResNeXt-101. The other parameters of the ST-CNN are initialized to random values. In terms of network optimization, we choose the stochastic gradient descent (SGD) with momentum of 0.9 and weight decay of 0.0005 as the optimization algorithm, and the number of iterations is set to 10 000. The initial value of the learning rate is 0.005. With the training, the learning rate is continuously adjusted through the poly strategy, in which the power value is 0.9. We resize all images and labels to 256×256, of which the type of defect image is png format and the type of label image is bmp format. In addition, in the training, we use mini batch size of four, which consists of two LIs and two UIs.

2) Inference Phase: In the test phase, we resize the test image to 256×256 and input it to the student network to directly obtain the prediction result (S_{p} in Fig. 3). In order to get more refined segmentation results, we apply a fully connected conditional random field (CRF) to postprocessing the results of network prediction.

Algorithm 1 Pseudo Code of the Proposed Method

Input: Labeled defect images x, Unlabeled defect images y.
Initialize: $base\ learning\ rate = 5 \times 10^{-3}, lr_decay = 0.9$
$batch_size = 4, item = 10000, \alpha = 10, \beta = 10$

1. For i =1 to item,do
2. $P_s, P_c, P_d = Student\ CNN(x)$
3. $Loss1 = \Phi_{BCE}(P_s, G_s) + \alpha \times \Phi_{BCE}(P_c, G_c)$ $+\beta \times \Phi_{BCE}(P_d, G_d)$
4. $S_s, S_c, S_d = Student\ CNN(y)$
5. $T_s, T_c, T_d = Teacher\ CNN(y)$
6. $Loss2 = \Phi_{MSE}(S_s, T_s) + \Phi_{MSE}(S_c, T_c)$ $+\Phi_{MSE}(S_d, T_d)$
7. $Loss = Loss1 + \lambda Loss2$
8. Backpropagation Loss
9. end for
10. Output: S_p (student network prediction image)

C. Evaluation Metrics

In order to better evaluate the model, we select four commonly used evaluation metric: balance error rate (BER), mean intersection over union (MIoU), mean pixel accuracy (MPA), and time. The BER can be given as follows:

$$\mathrm{BER} = \left(1 - \frac{1}{2}\left(\frac{N_{\mathrm{tp}}}{N_{\mathrm{p}}} + \frac{N_{\mathrm{tn}}}{N_{\mathrm{n}}}\right)\right) \times 100 \tag{9}$$

where N_{p}, N_{n}, N_{tp}, and N_{tn} are the number of defect, nondefect, true positives, and true negatives pixels of the defect images, respectively. The smaller the BER value, the better the detection effect.

MIoU and MPA are common metrics to measure the segmentation results. Since our network is a pixel-wise SegNet, these two indicators are used to measure the detection effect

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TABLE II
COMPARISON RESULTS OF DIFFERENT METHODS ON THREE DATASETS

Dataset	Evaluation	MT+ FCN	MT+ PSPNet	MT+ SegNet	MT+ UNet	MT+ DeepLabV3	MT+ BiSeNet	MT+ Unet++	MT+ STDC	Our
FID dataset	BER	25.18	26.18	33.34	19.66	23.71	20.37	20.55	22.00	**13.47**
	MIoU(%)	59.23	55.18	60.29	72.59	62.67	64.09	72.54	66.57	**82.57**
	MPA(%)	61.07	56.71	62.43	74.65	64.26	68.74	74.74	70.22	**91.85**
	Time(s)	0.039	0.041	0.040	0.039	**0.034**	0.072	0.071	0.100	0.059
AITEX dataset	BER	45.20	39.86	50.00	48.71	43.91	50.00	43.05	49.75	**19.82**
	MIoU(%)	51.09	51.49	50.40	50.57	51.04	50.40	51.11	50.46	**58.20**
	MPA(%)	53.46	53.66	52.65	52.95	53.29	52.65	53.48	52.74	**61.23**
	Time(s)	0.025	0.030	0.028	0.028	**0.023**	0.058	0.045	0.070	0.047
TILDA dataset	BER	45.97	44.66	44.38	44.86	43.98	50.00	39.28	46.13	**13.64**
	MIoU(%)	47.70	47.99	49.93	50.64	50.04	43.34	54.32	47.66	**79.14**
	MPA(%)	54.10	54.05	56.26	56.82	56.13	50.00	60.89	54.23	**87.77**
	Time(s)	0.031	0.029	0.029	0.028	**0.026**	0.061	0.066	0.091	0.048

of the network, as shown in the following:

$$\mathrm{MIoU} = \frac{1}{k}\sum_{i=1}^{k}\frac{P \cap G}{P \cup G} \tag{10}$$

where P and G represent the prediction and the ground truth, respectively

$$\mathrm{MPA} = \frac{1}{k+1}\sum_{i=0}^{k}\frac{p_{ii}}{\sum_{j=0}^{k} p_{ij}} \tag{11}$$

where p_{ii} represents the number of pixels correctly classified for each category (defects or nondefects), and p_{ij} represents the number of pixels predicted for that category. Both MIoU and MPA indicate that the larger the value, the better the detection effect.

D. Experimental Results and Discussion

In our experiment, the ratio of LIs and UIs in all training sets is set to 1:1. The number of training sets and test sets is shown in Table I. In order to better evaluate the defect detection performance of the network, many mainstream segmentation methods, such as FCN [6], pyramid scene parsing network (PSPNet) [35], SegNet [36], UNet [10], and DeepLabV3 [37], are embedded into the MT structure as semisupervised methods to compare with our method more fairly.

1) Detection Results on FID Dataset: Fig. 5 shows some fabric defect images and corresponding prediction results. The biggest challenge of this dataset is that there are multiple defects on each image, and the background of different defects is also quite different. It can be seen that our pixel-wise segmentation method can detect various defects better than other mainstream segmentation methods. For the comparison of evaluation indexes in Table II, our method can get better segmentation results. The value of BER is reduced to 13.47, the value of MIoU is improved to 82.57%, and the value of MPA is improved to 91.85%.

2) Detection Results on AITEX Dataset: Fig. 6 shows some defect images and corresponding detection results. The challenge of this dataset is that there are a large number of slender defects and small defects. It can be seen that the current mainstream segmentation methods cannot detect most defects of this dataset well. However, our method can detect defects with various shapes and has better detection results. In addition, it can be seen from the experimental results that our detection results are very similar to the ground truth. From the evaluation indicators in Table II, our method is much better than other mainstream segmentation methods in BER, MIoU, and MPA: the value of BER is reduced to 19.82, the value of MIoU is improved to 58.20%, and the value of MPA is improved to 61.23%.

TABLE III
COMPARISON OF MODEL SIZE AND MILLION PARAMETERS BETWEEN OUR MODEL AND CLASSICAL MODEL

Model	Model Size (MB)	Million Parameters
AlexNet [5]	>200	60
VGG 16 [38]	>500	138
Inception V3 [39]	90-100	23.2
ResNet-152 [29]	230	60
Our Network	169.24	44.14

3) Detection Results on TILDA Dataset: Fig. 7 shows some defect images and corresponding defect detection results. The challenge of this dataset is that many defects and backgrounds are similar. It can be seen that the current mainstream segmentation methods can detect large defects better (as shown in the second column of Fig. 7). However, when the defect is similar to the background, it cannot be detected better (as shown in the fourth column of Fig. 7). It can be seen that our method can detect all kinds of defects better. From the evaluation indicators in Table II, our method has greatly improved in several evaluation indicators: the value of BER is reduced to 13.64, the value of MIoU is improved to 79.14%, and the value of MPA is improved to 87.77%.

4) Model Efficiency Analysis: We compare our model with the classical CNN in terms of model size and million parameters, as shown in Table III. It can be seen that our model is of medium size. It is slightly larger than the lightweight Inception V3 model, but compared with other mainstream CNN backbone networks, our model is of normal size in model size and million parameters. In addition, according to

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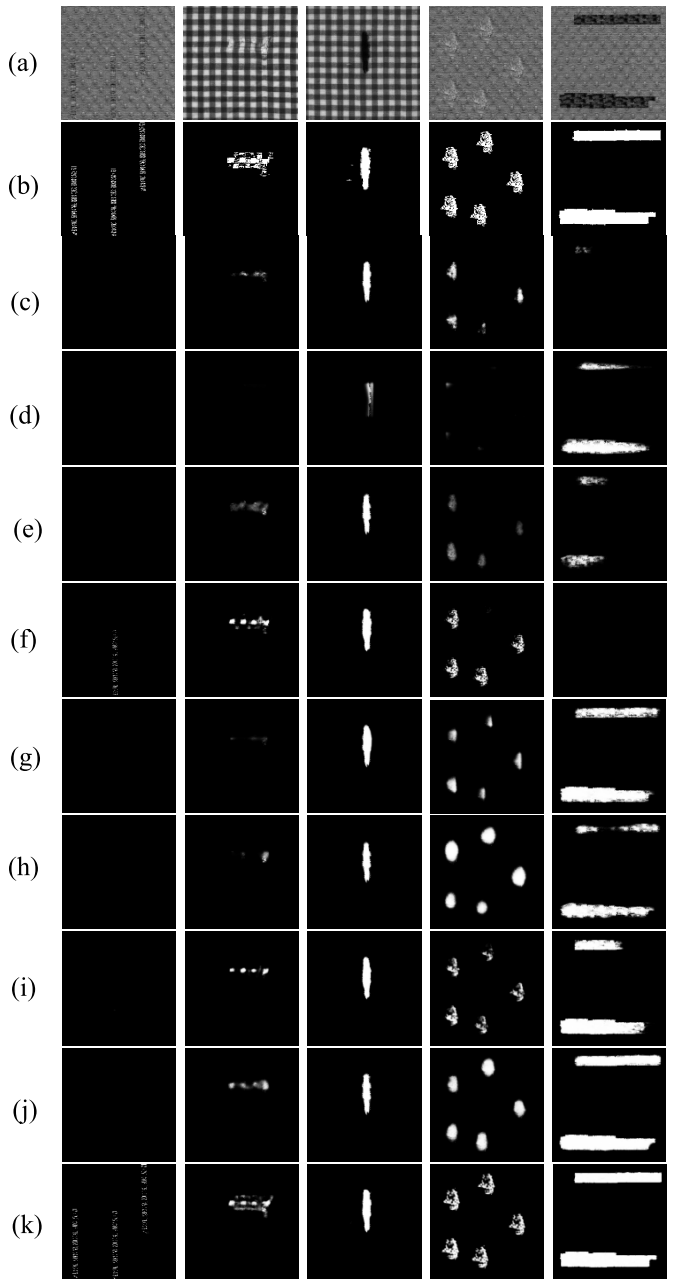


Fig. 5. Comparison of detection results on FID dataset. (a) Original defect image. (b) Ground truth. (c) MT + FCN [6]. (d) MT + PSPNet [35]. (e) MT + SegNet [36]. (f) MT + UNet [10]. (g) MT + DeepLabV3 [37]. (h) MT + BiSeNet [11]. (i) MT + UNet++ [12]. (j) MT + STDC [13]. (k) Our network.

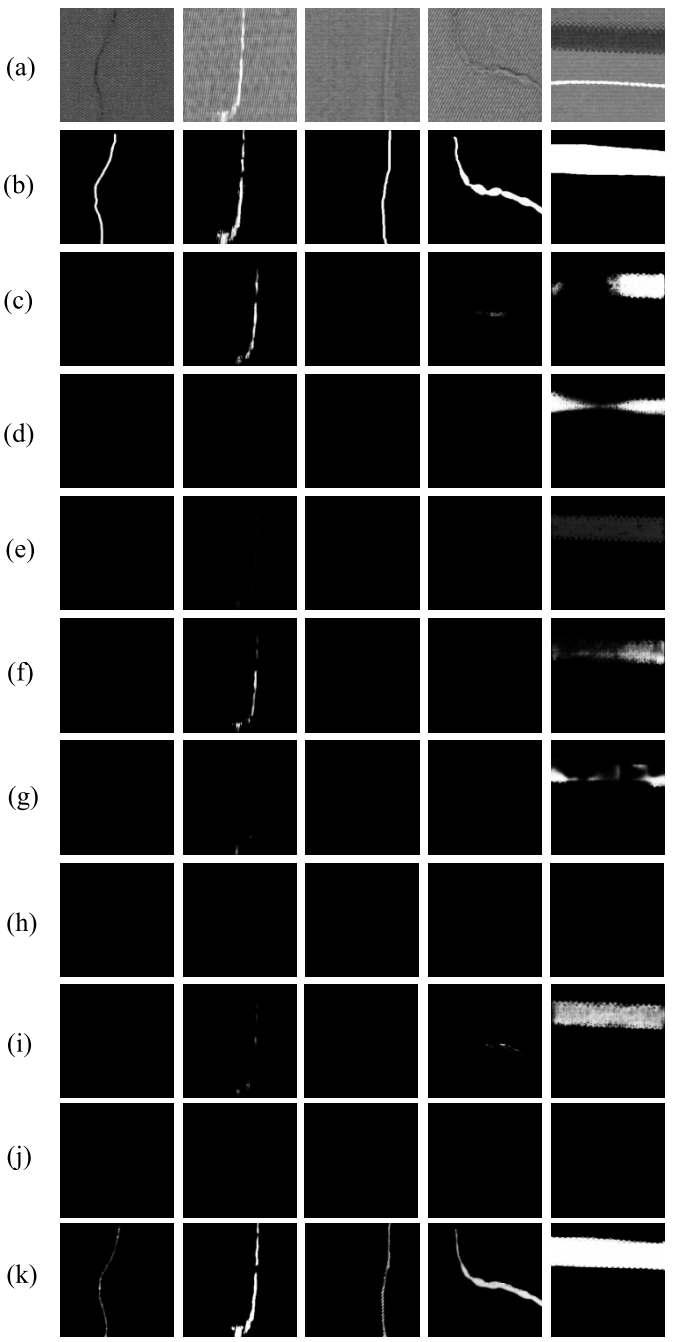


Fig. 6. Comparison of detection results on AITEX dataset. (a) Original defect image. (b) Ground truth. (c) MT + FCN [6]. (d) MT + PSPNet [35]. (e) MT + SegNet [36]. (f) MT + UNet [10]. (g) MT + DeepLabV3 [37]. (h) MT + BiSeNet [11]. (i) MT + UNet++ [12]. (j) MT + STDC [13]. (k) Our network.

the datasets in Table II, we also give the running time of a single image of the model. From Table II, we can see that the running time of our method is about 0.048 s per image, which is equivalent to about 20 fabric defect images can be detected per second, and the detection speed can meet certain industrial applications.

5) Study on Noisy Data: In the actual industrial scene, the defect image will be affected by noise due to changes in environment, illumination, and other factors. To verify the robustness of the method against noisy samples, we first have to generate noisy samples and then test the effectiveness of defect detection on these noisy samples. As the method in [40], we use the FID dataset to generate the degraded dataset by blurring the samples in FID dataset with a Gaussian function and then polluting them with salt and pepper noise. The used Gaussian function is expressed as follows:

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{(x-m/2)^2+(y-n/2)^2}{2\sigma^2}} \tag{12}$$

where (m, n) represents the size of 2-D filter template and σ represents the standard deviation of normal distribution. In our experiment, different sizes (m, n) and σ are selected

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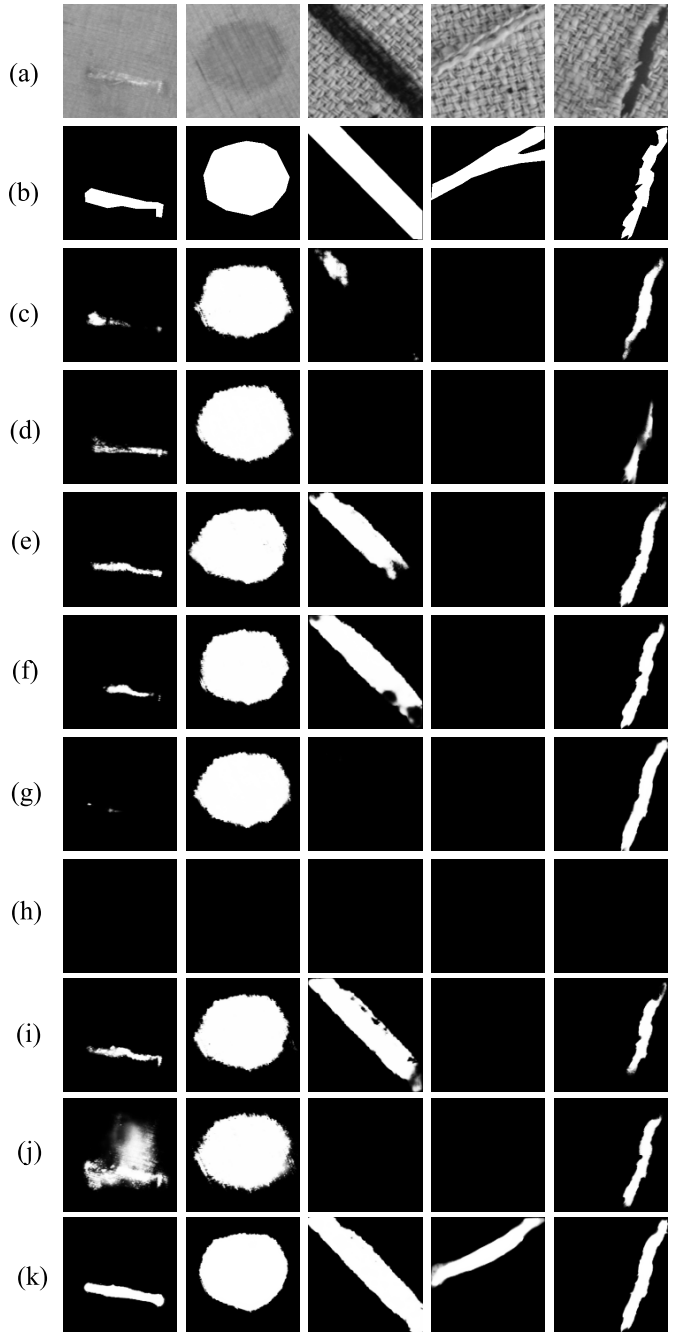


Fig. 7. Comparison of detection results on TILDA dataset. (a) Original defect image. (b) Ground truth. (c) MT + FCN [6]. (d) MT + PSPNet [35]. (e) MT + SegNet [36]. (f) MT + UNet [10]. (g) MT + DeepLabV3 [37]. (h) MT + BiSeNet [11]. (i) MT + UNet++ [12]. (j) MT + STDC [13]. (k) Our network.

to produce different degrees of blurred samples, and the parameter of the salt and pepper noise is set as 0.1.

Fig. 8 illustrates four groups of defective samples blurred with different parameters. From Fig. 8, it can be seen that the proposed method can detect defects well under general degradation conditions. When the defect image has serious blur and noise, as shown in Fig. 8(f), although the detection effect of the proposed method is reduced to a certain extent, it still achieves acceptable results except the result in the fourth row of Fig. 8(f).

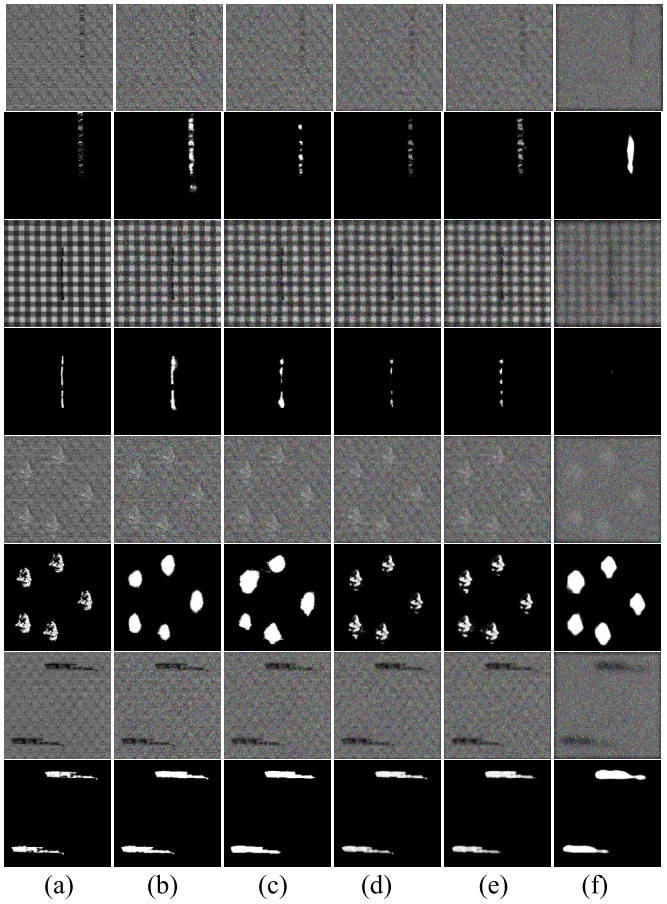


Fig. 8. Four groups of fabric defect detection results under different degradation conditions. (a) No degradation. (b) $(m, n) = (2, 2), \sigma = 2$. (c) $(m, n) = (4, 4), \sigma = 2$. (d) $(m, n) = (6, 6), \sigma = 2$. (e) $(m, n) = (8, 8), \sigma = 2$. (f) $(m, n) = (20, 20), \sigma = 10$.

E. Ablative Study

1) Network Structure Analysis: In the design of network structure, the main innovation of this model is that on the one hand, the defect contour and DM are used as structural prior information to assist the pixel-wise segmentation of the network; on the other hand, an effective ST-CNN is designed, and effective feature fusion is used in ST-CNN to increase the diversity of features, which are helpful to extract richer defect information. In order to verify the effectiveness of the network structure, we conduct ablation experiments for different parts of the network. It includes ablation study for feature fusion, ablation study for contour information, and ablation study for DM. The analysis results are shown in Table IV, where "MT" is the basic network without contour and DM information, "+" means to increase the structure on the basic network, and "−" means to remove the structure on the basic network. Therefore, "MT-FF" means removing the feature fusion structure, "MT + DC" means adding defect contour information, and "MT + DC + DD" means adding defect contour and DM. All experimental results are based on the TILDA dataset.

First, on the basis of the basic network, we remove the feature fusion part of the design, which can significantly reduce the network performance, which shows that feature fusion can

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TABLE IV
ABLATION ANALYSIS ON TILDA DATASET

	Method	BER	MIoU(%)	MPA(%)
Network structure analysis	MT-FF	50.00	43.34%	50.00%
	MT	15.27	76.81%	84.48%
	MT+DC	14.14	78.23%	85.31%
	MT+DC+DD	13.64	79.14%	87.77%
Loss function analysis	SL	13.98	77.62%	84.86%
	ML	13.64	79.14%	87.77%

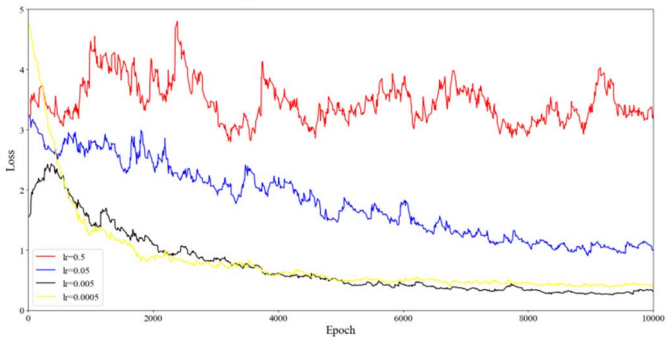


Fig. 9. Curve of loss with epoch under different learning rates.

indeed extract more effective features, which is conducive to fabric defect detection. Then, on the basis of the basic network, the contour information is added, and the detection effect is improved. After further adding DM information, the detection effect reaches the best. The experimental results of ablation study show that the contour information and DM of defects can indeed provide structural prior information to the defect region to assist in better network segmentation.

2) Loss Function Analysis: In the design of loss function, the main innovation is to combine the defect segmentation loss, defect contour loss, and defect DM loss as multitask loss. According to the characteristics of MT network structure, multitask supervised loss function and multitask consistency loss function are constructed. In order to verify the effectiveness of multitask loss function, we conduct ablation experiments for multitask loss function and single-task loss function, as shown in Table IV, where "SL" represents single-task loss function, that is, only segmentation loss, no contour loss, and DM loss. "ML" represents our proposed multitask loss function. The experimental results show that the multitask loss function can indeed improve the segmentation rate of fabric defect detection.

3) Hyper-Parameters Analysis: In our experiment, learning rate and epoch are important hyperparameters. In order to obtain good hyperparameters, we experimentally analyze the curve of loss with epoch under different learning rates, as shown in Fig. 9. In the selection of learning rate, we set the learning rate in the range of $10^{-1} - 10^{-4}$. It can be seen from Fig. 9 that when the learning rate $\mathrm{lr} = 0.005$, the value of loss can continue to decline; when the learning rate $\mathrm{lr} > 0.005$, the loss value is in a state of constant oscillation; and when the learning rate $\mathrm{lr} < 0.005$, the loss value did not decrease to the minimum value. Therefore, we choose $\mathrm{lr} = 0.005$ as our best parameter. In the selection of epoch, it can be seen from Fig. 9 that when the epoch is about 10 000, the loss value can be reduced to the minimum and kept in a stable state.

V. CONCLUSION

In this article, a pixel-wise semisupervised fabric defect detection method is proposed. We mainly develop on the basis of semisupervised framework MT network. We first establish a multitask detection network (ST-CNN), which use the defect contour and DM as structural prior information to assist the network in better defect segmentation. Then, we take the multitask detection network as the student network and the teacher network to predict the labeled data and the unlabeled data, in which the multitask supervised loss is used for the labeled data and the multitask consistency loss is used for the unlabeled data to optimize the network. Finally, the way of deep supervision is used to promote network learning. In conclusion, the experiments on three public fabric defect detection show the strengths of the model. In qualitative analysis, it can achieve better detection results. In quantitative analysis, the relevant evaluating indicator exceeds the current mainstream pixel-wise segmentation methods and has a better application prospect. However, in our experiment, the unit detection time of the network is higher than the mainstream SegNet. Although the detection speed can meet the general industrial detection requirements, the speed needs to be further improved for textile workshops with high-speed detection scene. It is our future work to compress the model to obtain a lightweight model under the condition of ensuring high detection accuracy.

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